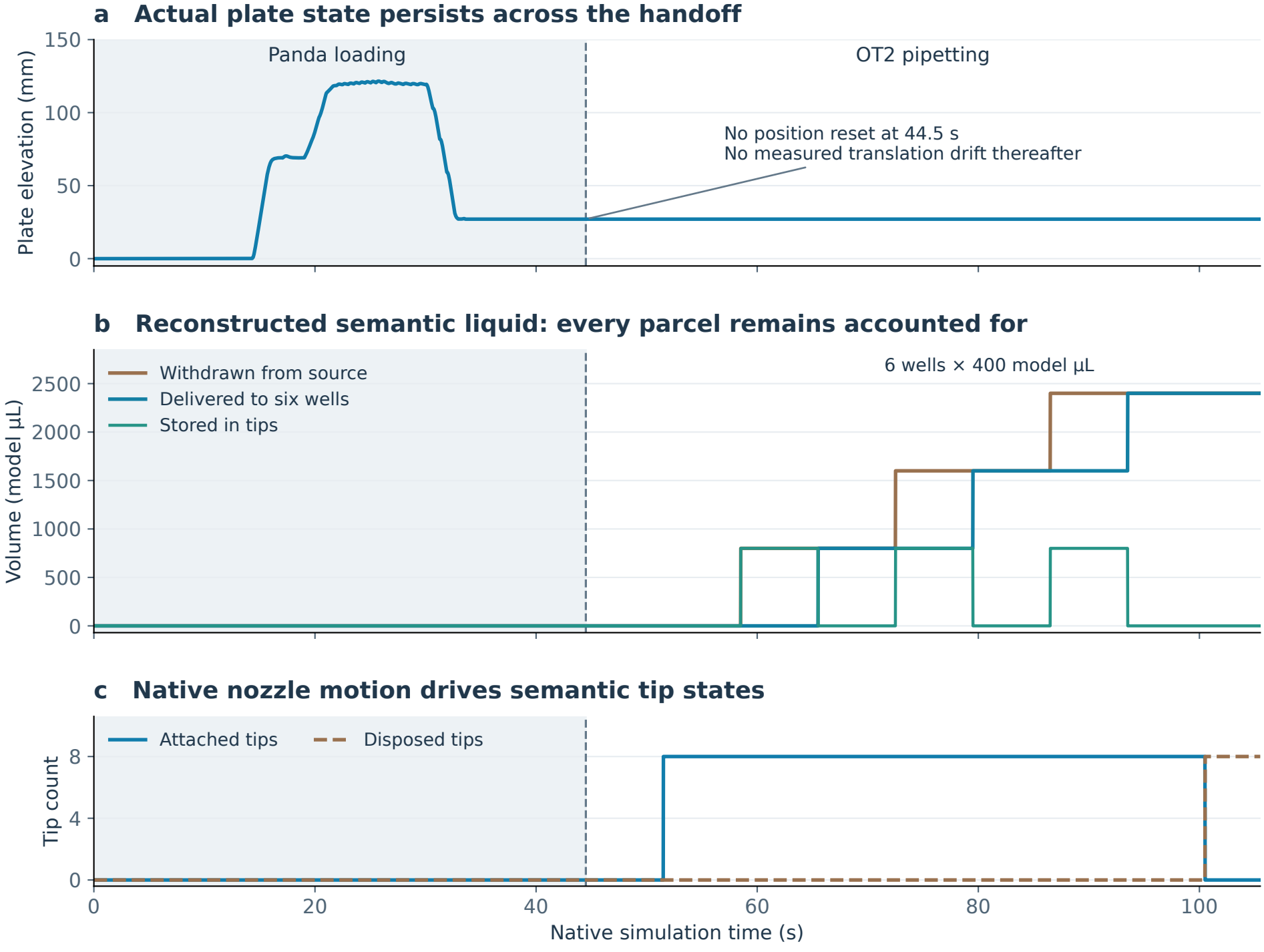
One physical scene: plate loading → pipetting

105.5 s · 63,300 native steps · 64 geometry-gated semantic events



One independently checked positive trial; the continuous no-aspiration control is separate.
Local support / grip overlay and bounded door-holding torque; simulator-ground-truth control.
Tip attachment and finite-volume liquid are explicit semantics, not friction-fit or calibrated particle-fluid physics.